

Shen Zhen MTC Co., LTD

TEST REPORT

SCOPE OF WORK

Radio Frequency Testing—MHDV4080-T9503-S2,
MHDV4080-T9503, MHDV40****- T9503-##,
MHDV40****- T9503, (*, # can from 0 to 9, A to Z or
blank), 2T-C40HD2225EB, 2T-C40HD2225KB, 2T-
C40HD2725KB, 2T-C40HD2725EB, 2T-C40HD3225EB,
2T-C40HD3225KB, 2T-C40HD2x25EB, 2T-C40HD2x25KB,
-*40***(* can from 0 to 9,A to Z or blank),
PR40F2462VKB, F40AR5G, LT-40CR350, L40RFE25

REPORT NUMBER

260508019SZN-004

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ETSI EN300328_c
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RADIO COMMUNICATIONS TESTING REPORT

Shen Zhen MTC Co., LTD

MHDV4080-T9503-S2, MHDV4080-T9503, MHDV40****-T9503-##, MHDV40****-T9503, (*, # can from 0 to 9, A to Z or blank), 2T-C40HD2225EB, 2T-C40HD2225KB, 2T-C40HD2725KB, 2T-C40HD2725EB, 2T-C40HD3225EB, 2T-C40HD3225KB, 2T-C40HD2x25EB, 2T-C40HD2x25KB, **-*40*****(* can from 0 to 9,A to Z or blank), PR40F2462VKB, F40AR5G, LT-40CR350, L40RFE25

TV/LED TV

2.4GHz Transceiver

Test Report: 260508019SZN-004

Report Prepared and Checked:	Bruce Zheng Project Engineer	
Report Approved By:	Rode Liu Project Engineer	
Date:	20 May 2026	

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RADIO PERFORMANCE MEASUREMENTS

RESULT SUMMARY

Requirements	ETSI EN 300 328		Condition	Compliance
	Technical requirements	Test Specification		
	Clause Number			
RF Output Power	4.3.2.2	5.4.2	Only for modulations other than FHSS Only for non-adaptive equipment and RF Output Power>10dBm	Complied
Power Spectral Density	4.3.2.3	5.4.3		Complied
Duty cycle, Tx-Sequence, Tx-gap	4.3.2.4	5.4.2		N/A
Medium Utilisation	4.3.2.5	5.4.2	Only for non-adaptive equipment and RF Output Power>10dBm	N/A
Adaptivity	4.3.2.6	5.4.6	Only for adaptive equipment and RF Output Power>10dBm	Complied
Occupied Channel Bandwidth	4.3.2.7	5.4.7		Complied
Transmitter unwanted emission in the OOB domain	4.3.2.8	5.4.8		Complied
Transmitter unwanted emission in the spurious domain	4.3.2.9	5.4.9		Complied
Receiver spurious emissions	4.3.2.10	5.4.10		Complied
Receiver Blocking	4.3.2.11	5.4.11		Complied
Geo-location capability	4.3.2.12	N/A	Only applies to equipment with geo-location capability	N/A
Technical requirements for FHSS modulation	4.3.1	5.4	Only for FHSS	N/A
When determining the test conclusion, the Measurement Uncertainty of test has been considered. Note: N/A means Not applicable.				

**EQUIPMENT UNDER TEST (EUT)
INFORMATION**

Applicant: Shen Zhen MTC Co., LTD
MTC Industry Park, 1st Lilang Road, Xialilang community, Nanwan
Longgang district, Shenzhen, China

Description of EUT: TV/LED TV

Type Number (s): MHDV4080-T9503-S2

Brand Name(s): AMTC, SHARP, Polariod, VITADO, JVC, Logik

Serial Number (s): Not Labelled

Equipment Received: 14 January 2025

Test Date (s): 14 January 2025 to 12 February 2025

WiFi Version: 802.11b,
802.11g,
802.11n-HT20 ,
802.11n-HT40

Test Site and Location: Intertek Testing Services Shenzhen Ltd.
No.101&201, Building B, No.308, Wuhe Avenue, Zhangkengjing, Guanhu
Subdistrict, Longhua District, Shenzhen, Guangdong, China

Test Specification (s): ETSI EN 300 328 V2.2.2 (2019-07)

CONTENTS

EXHIBIT 1 GENERAL DESCRIPTION	5
1 Introduction	6
2 Test Specification	6
EXHIBIT 2 TEST RESULT OF RADIO PERFORMANCE MEASUREMENTS	8
3 Transmitter unwanted emissions in the spurious domain	9
4 Receiver spurious emission	15
EXHIBIT 3 PHOTOS OF EUT	21
5 EUT Photos	22

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EXHIBIT 1

GENERAL DESCRIPTION

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1 INTRODUCTION

Intertek Testing Services Shenzhen Limited has tested the TV/LED TV MHDV4080-T9503-S2. The sample was tested to the relevant performance specification published by the European Telecommunications Standards Institute. This report contains the results of these tests and is submitted Shen Zhen MTC Co., LTD as the final test results.

The equipment under test (EUT) is a TV/LED TV with WIFI function; The EUT was powered 220-240V~, 50/60Hz, 70W or 90W, the detail please see the user manual.

The Model MHDV4080-T9503, MHDV40****- T9503-##, MHDV40****- T9503, (*, # can from 0 to 9, A to Z or blank), 2T-C40HD2225EB, 2T-C40HD2225KB, 2T-C40HD2725KB, 2T-C40HD2725EB, 2T-C40HD3225EB, 2T-C40HD3225KB, 2T-C40HD2x25EB, 2T-C40HD2x25KB, **..*40*****(* can from 0 to 9,A to Z or blank), PR40F2462VKB, F40AR5G, LT-40CR350, L40RFE25 are the same as the Model: MHDV4080-T9503-S2 in hardware aspect. The difference in model number serves as marketing strategy.

This report bases on the previous report with report number 251219032SZN-002 Dated 09 January 2026 (original signature by Bruce Zheng, Rode Liu on file). Owing to adding the models, no test was required after evaluation.

Due to the use of the same wireless module, Intertek only conducted the Radiated Spurious Emissions test. The results of other test items were cited from the module report CN23DLMM 001 Date of issue is 27 October 2023 (original signature by Breeze Jiang, Lin Lin on file).

The production units are required to conform to the initial sample as received when the units are placed on the market.

2 TEST SPECIFICATION

2.1 RELEVANT PERFORMANCE SPECIFICATION

The relevant performance specifications for Shen Zhen MTC Co., LTD MHDV4080-T9503-S2 are ETSI EN 300 328 V2.2.2 (2019-07). The harmonised standards are ETSI EN 300 328 V2.2.2 (2019-07).

The tests performed are those required to demonstrate compliance with the technical specifications and the essential requirements of Article 3.2 of the Radio Equipment Directive (2014/53/EU) - RED for regulatory purposes.

2.2 TEST ENVIRONMENT

The tests were performed in the Radio communications and Electromagnetic Compatibility Test Facility at Intertek Testing Services Shenzhen Limited. The sample was subjected to the ambient conditions in the laboratory and indoor test site except during tests at extremes of temperatures and the Radiated Emissions Tests. The temperature and relative humidity recorded during the period of each test are given in the results.

2.3 CONFIGURATION OF TEST SAMPLE

The test sample consisted of one transceiver and under manufacturer's specified test mode, only the worst case data were reported.

During testing channel & power controlling software provide by the applicant was used to control the operation channel as well as the output power level. The RF output power selection is for the setting of RF output power expected by the applicant and is going to be fixed on the firmware of the final end product.

2.4 TEST POWER SOURCES

The TV/LED TV is intended to operate from 220-240V~, 50/60Hz, 70W or 90W. The test power source voltages were:

Nominal test voltage

AC 230V

2.5 TEST FREQUENCIES

For 802.11b/g and n-HT20 the sample supplied operated on 13 channels, nominally at 2.412 - 2.472 GHz for Transceiver. The channel is separated by 5 MHz channel spacing. The tests were carried out on channel 1, 7 and 13 channels of the frequency of the alignment range.

For 802.11n-HT40 the sample supplied operated on 9 channels, nominally at 2.422 - 2.462 GHz for Transceiver. The channel is separated by 5 MHz channel spacing. The tests were carried out on channel 3, 7 and 11 channels of the frequency of the alignment range.

2.6 GENERAL REQUIREMENTS**2.6.1 MODULATION**

1. Constant envelope digital frequency modulation is used with CCK / DSSS / OFDM technique,
2. The product is a TV/LED TV with WiFi IEEE 802.11b/g/n-HT20/n-HT40 radio functions.

2.6.2 ANTENNA

The antenna used in Transceiver is permanent Integrate antenna.

Antenna Gain: 1.72dBi

2.6.3 Test Condition

Manufacturer's declared operating temperature: -10°C to +55°C.

2.7 MEASUREMENT UNCERTAINTY

All measurement uncertainties stated in this report are estimated to a 95% confidence level.

2.8 SUPPORT EQUIPMENT – radio performance measurements

1. Laptop (Dell 5420) (Provided by Intertek)
2. USB Cable (Detachable, un-shielded, 120cm) (Provided by Applicant)
3. Test Software (Provided by Applicant) was installed on the EUT

EXHIBIT 2

**TEST RESULT
OF
RADIO PERFORMANCE MEASUREMENTS**

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3 TRANSMITTER UNWANTED EMISSIONS IN THE SPURIOUS DOMAIN

3.1 TEST METHOD AND SUMMARY

Basic Standard:	ETSI EN 300 328 V2.2.2 (2019-07)
Clause:	5.4.9
Test method	Radiated measurements

3.2 EQUIPMENT LIST

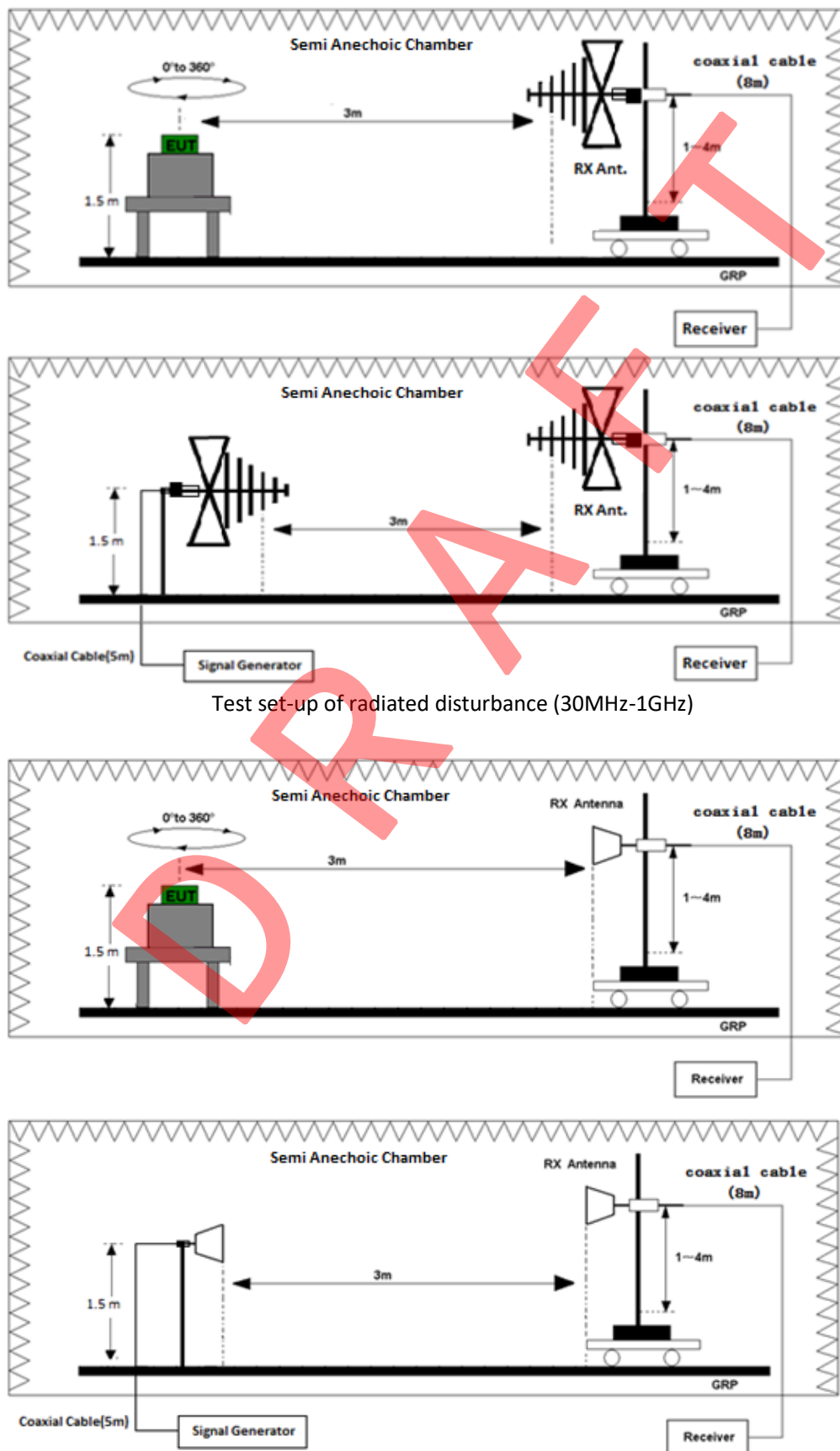
Equip No.	Description	Manufacturer	Model No.	Cal. Date	Due Date
SZ185-03	EMI Receiver	R&S	ESR7	2024-04-23	2025-04-23
SZ056-06	Signal Analyzer	R&S	FSV40	2024-12-06	2025-12-06
SZ061-13	Biconilog Antenna	ETS	3142E	2022-07-13	2025-07-13
SZ061-08	Horn Antenna	ETS	3115	2024-09-13	2027-09-13
SZ181-08	Microwave System Amplifier	keysight	83017A	2024-07-29	2025-07-29
SZ188-05	Anechoic Chamber	ETS	FACT 3-2.0	2021-05-25	2026-05-25
SZ062-40	RF Cable	Talent Microwave	A50-3.5M3.5M-4.5M	2024-09-30	2025-09-30
SZ062-41	RF Cable	Talent Microwave	A50-3.5M3.5M-8M	2024-09-30	2025-09-30
SZ062-34	RF Cable	Talent Microwave	A50-3.5M3.5M-1M	2024-09-30	2025-09-30

3.3 LIMIT

The transmitter unwanted emissions in the spurious domain shall not exceed the values.

Frequency Range	Maximum power	Bandwidth
30 MHz to 47 MHz	-36 dBm	100 kHz
47 MHz to 74 MHz	-54 dBm	100 kHz
74 MHz to 87,5 MHz	-36 dBm	100 kHz
87,5 MHz to 118 MHz	-54 dBm	100 kHz
118 MHz to 174 MHz	-36 dBm	100 kHz
174 MHz to 230 MHz	-54 dBm	100 kHz
230 MHz to 470 MHz	-36 dBm	100 kHz
470 MHz to 694 MHz	-54 dBm	100 kHz
694 MHz to 1 GHz	-36 dBm	100 kHz
1 GHz to 12,75 GHz	-30 dBm	1 MHz

3.4 TEST SETUP DIAGRAM:



Test set-up of radiated disturbance (30MHz-1GHz)

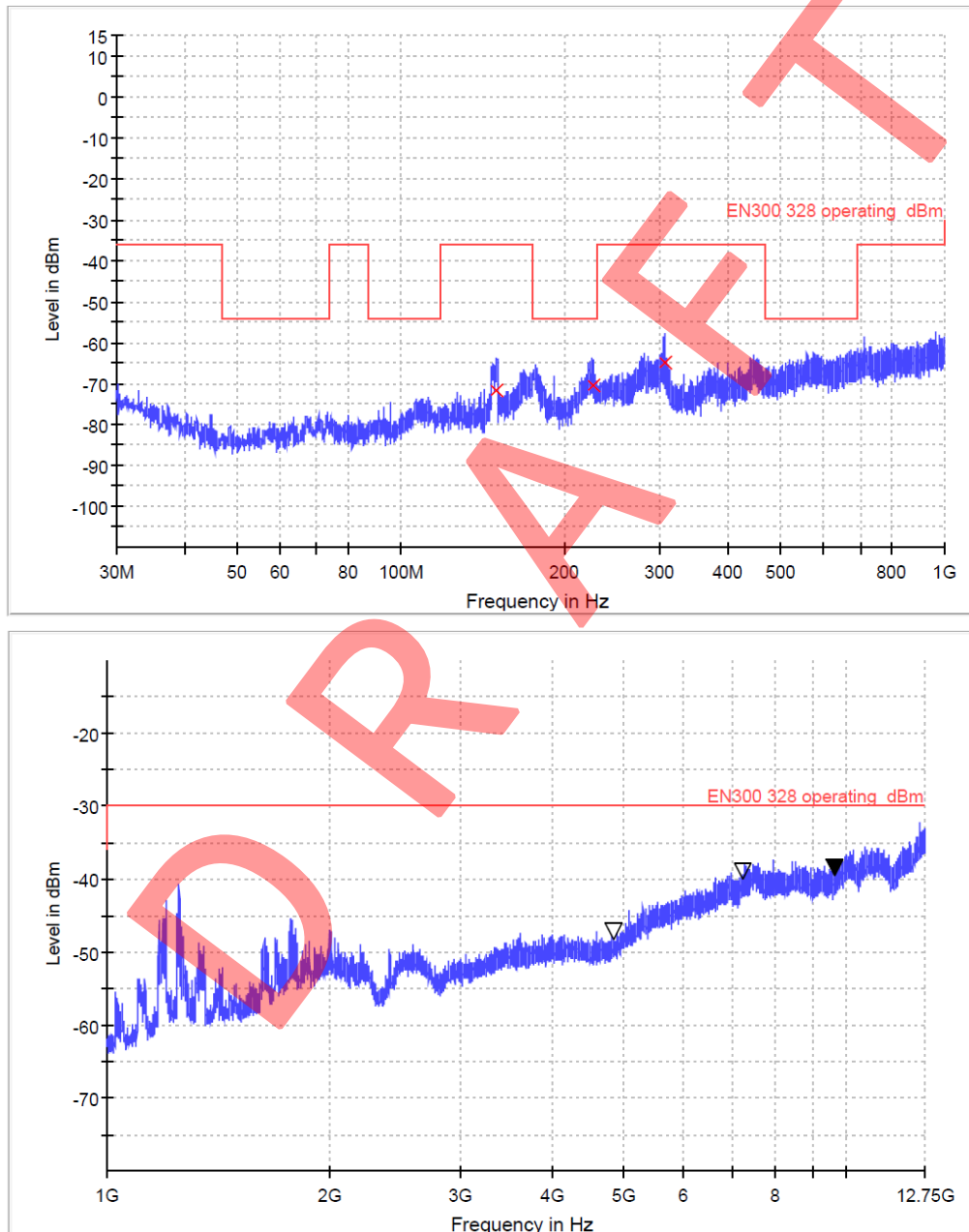
Test set-up of radiated disturbance (above 1GHz)

3.5 TEST RESULT

(802.11b) - Worst case: ANT 1

Channel 1

Antenna Polarization Horizontal:



Limit and Margin

Frequency (MHz)	RMS (dBm)	Meas. Time (ms)	Pol	Margin - RMS (dB)	Limit - RMS (dBm)
225.649000	-70.5	1000.0	H	16.5	-54.0
7236.000000	-39.7	1000.0	H	9.7	-30.0
9648.000000	-39.3	1000.0	H	9.3	-30.0

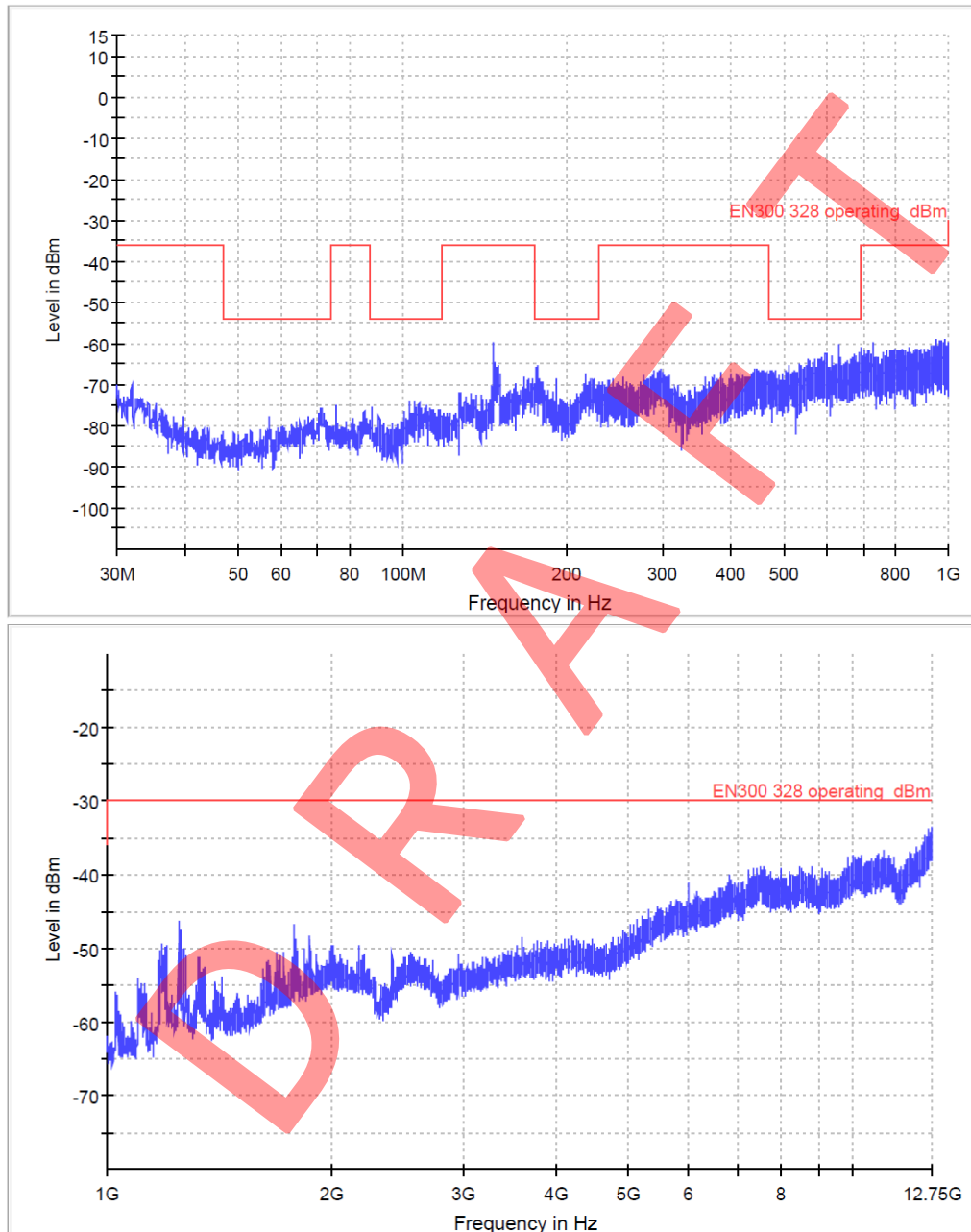
☐ No emissions significantly above equipment noise floor.

Notes:

1. The test frequency range is 30MHz to 12.75GHz.
2. Other emissions found were at least 10 dB below the limit.
3. Measurement Uncertainty: ± 4.8 dB for frequency range 30MHz to 1GHz and ± 6.0 dB for frequency range 1GHz to 12.75GHz.

Channel 13

Antenna Polarization Horizontal:



Limit and Margin

Frequency (MHz)	RMS (dBm)	Meas. Time (ms)	Pol	Margin - RMS (dB)	Limit - RMS (dBm)
147.325500	-63.5	1000.0	H	9.5	-54.0
7416.000000	-38.6	1000.0	H	8.6	-30.0
9888.000000	-39.2	1000.0	H	9.2	-30.0

☐ No emissions significantly above equipment noise floor.

Notes:

1. The test frequency range is 30MHz to 12.75GHz.
2. Other emissions found were at least 10 dB below the limit.
3. Measurement Uncertainty: ± 4.8 dB for frequency range 30MHz to 1GHz and ± 6.0 dB for frequency range 1GHz to 12.75GHz.

4 RECEIVER SPURIOUS EMISSION

4.1 TEST METHOD AND SUMMARY

Basic Standard:	ETSI EN 300 328 V2.2.2 (2019-07)
Clause:	5.4.10
Test method	Radiated measurements
Test Mode	Receive-only mode

4.2 EQUIPMENT LIST

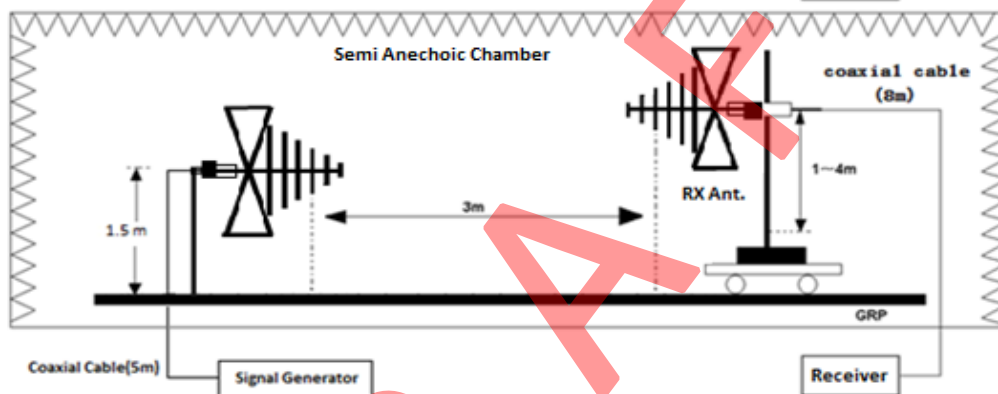
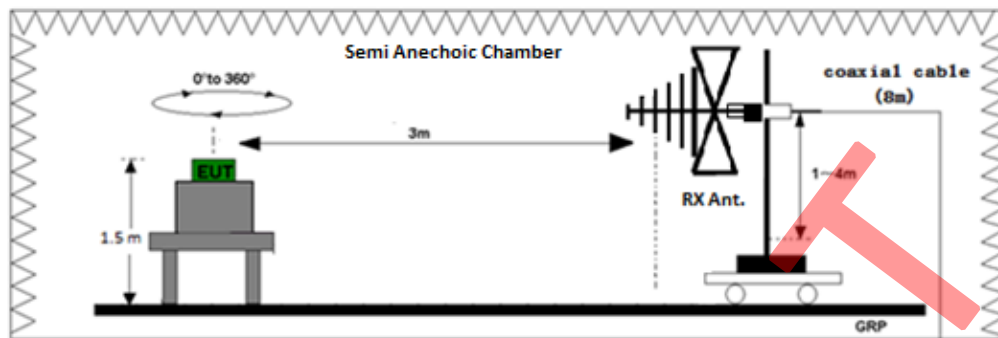
Equip No.	Description	Manufacturer	Model No.	Cal. Date	Due Date
SZ185-03	EMI Receiver	R&S	ESR7	2024-04-23	2025-04-23
SZ056-06	Signal Analyzer	R&S	FSV40	2024-12-06	2025-12-06
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SZ062-40	RF Cable	Talent Microwave	A50-3.5M3.5M-4.5M	2024-09-30	2025-09-30
SZ062-41	RF Cable	Talent Microwave	A50-3.5M3.5M-8M	2024-09-30	2025-09-30
SZ062-34	RF Cable	Talent Microwave	A50-3.5M3.5M-1M	2024-09-30	2025-09-30

4.3 LIMIT

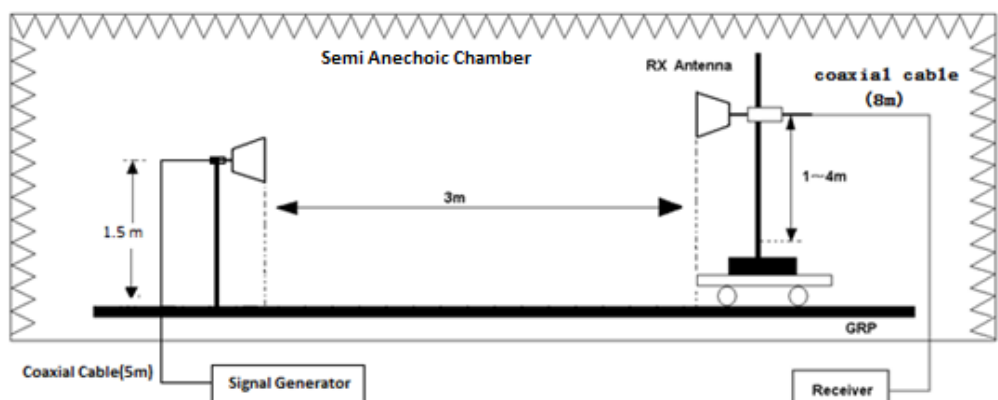
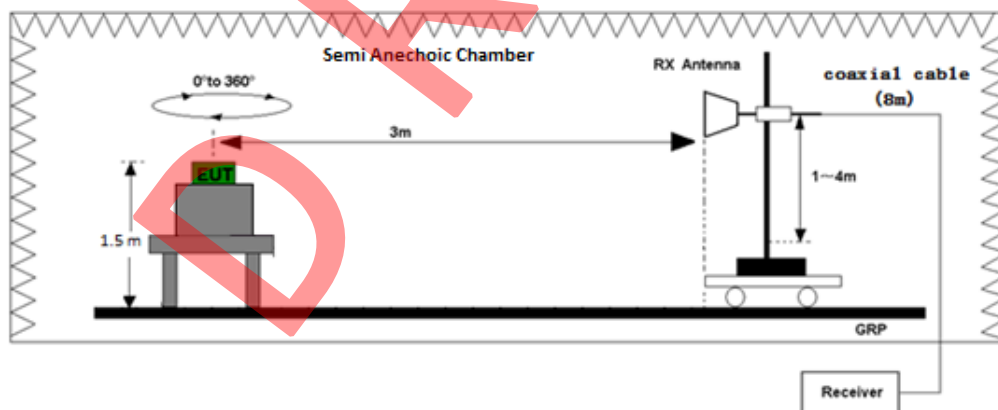
The transmitter unwanted emissions in the spurious domain shall not exceed the values.

Frequency Range	Maximum power	Bandwidth
30 MHz to 1 GHz	-57 dBm	100 kHz
1 GHz to 12.75 GHz	-47 dBm	1 MHz

4.4 TEST SETUP DIAGRAM:



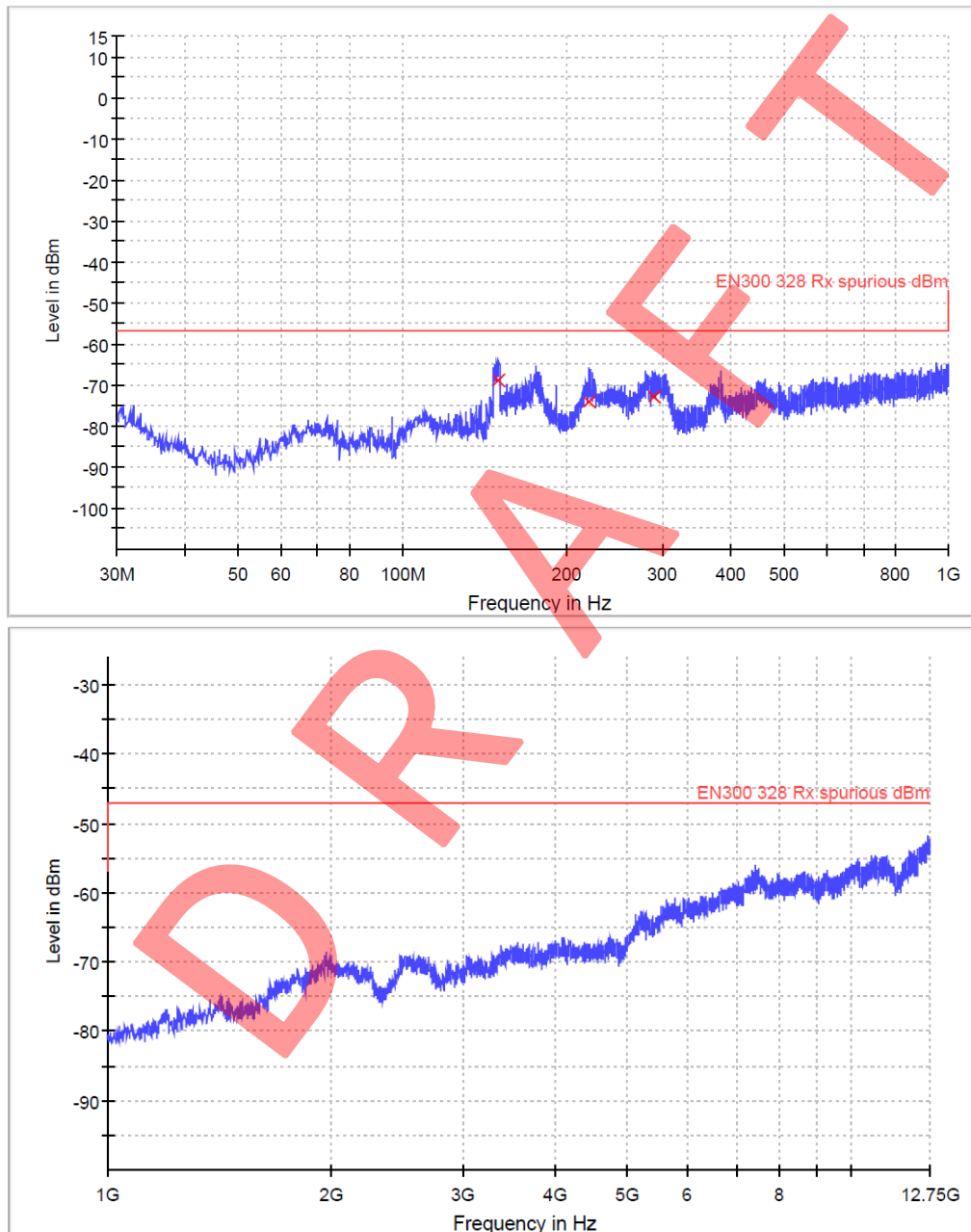
Test set-up of radiated disturbance (30MHz-1GHz)



Test set-up of radiated disturbance (above 1GHz)

4.5 TEST RESULT

(802.11b) - Worst case: ANT 1
Channel 1



Limit and Margin

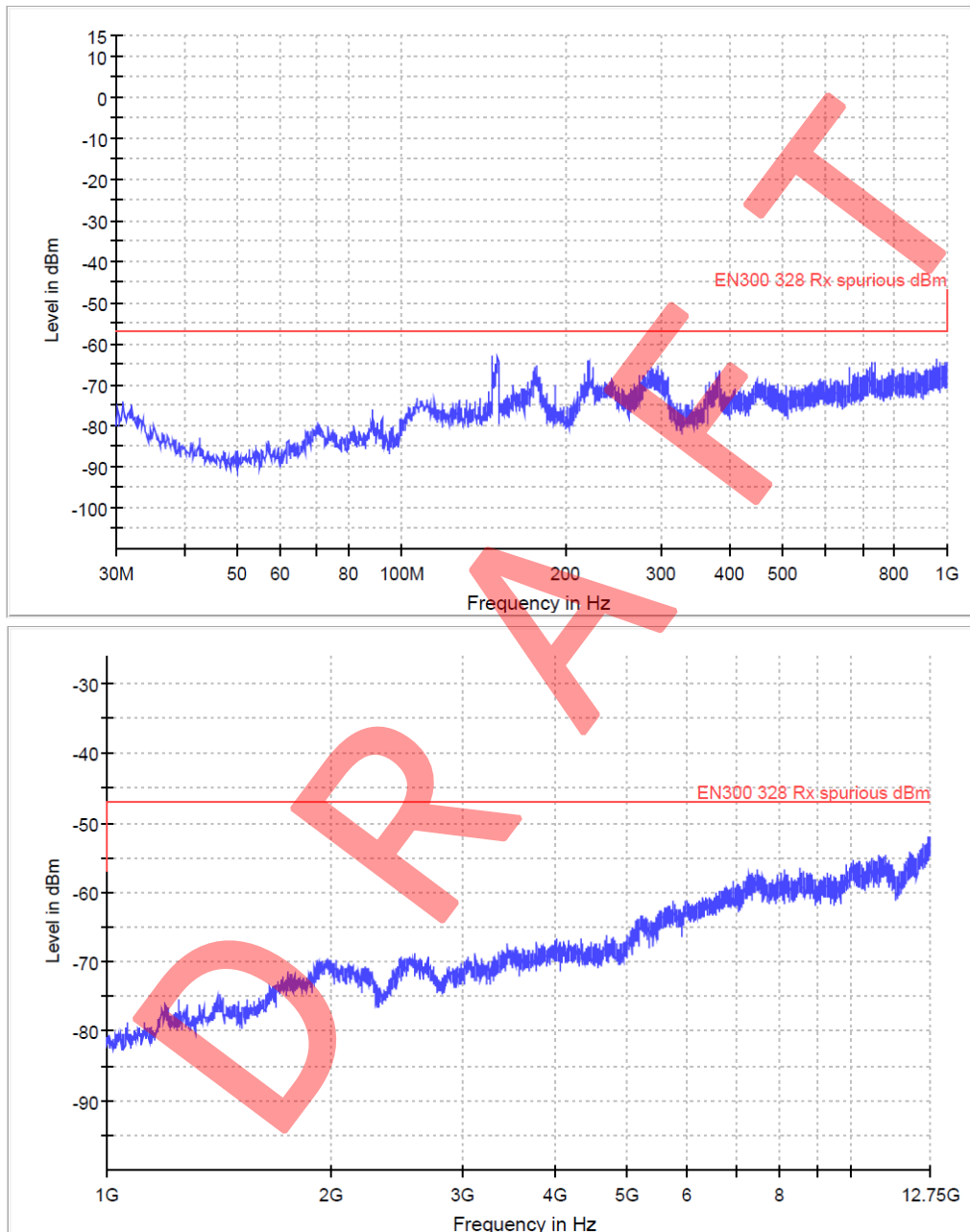
Frequency (MHz)	RMS (dBm)	Meas. Time (ms)	Pol	Margin - RMS (dB)	Limit - RMS (dBm)
150.401250	-68.7	1000.0	H	11.7	-57.00
219.998750	-73.9	1000.0	H	16.9	-57.00
288.383750	-72.9	1000.0	H	15.9	-57.00

☐ No emissions significantly above equipment noise floor.

Notes:

1. The test frequency range is 30MHz to 12.75GHz.
2. Other emissions found were at least 10 dB below the limit.
3. Measurement Uncertainty: ± 4.8 dB for frequency range 30MHz to 1GHz and ± 6.0 dB for frequency range 1GHz to 12.75GHz.

Channel 13



Limit and Margin

Frequency (MHz)	RMS (dBm)	Meas. Time (ms)	Pol	Margin - RMS (dB)	Limit - RMS (dBm)
149.231250	-69.7	1000.0	H	12.7	-57.00
220.963750	-72.3	1000.0	H	15.3	-57.00
302.383750	-74.1	1000.0	H	17.1	-57.00

☐ No emissions significantly above equipment noise floor.

Notes:

1. The test frequency range is 30MHz to 12.75GHz.
2. Other emissions found were at least 10 dB below the limit.
3. Measurement Uncertainty: ± 4.8 dB for frequency range 30MHz to 1GHz and ± 6.0 dB for frequency range 1GHz to 12.75GHz.

EXHIBIT 3

PHOTOS OF EUT

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5. PHOTOS OF TEST SET-UP

Please refer to Annex of EUT Photos_260508019SZN.

***** End of Report *****

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